Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 4: Fourier series

How Topic 4 builds on previous topics:

- In Topic 2, we used power series to solve ODEs with smooth solutions and within a radius of convergence
- In Topic 3, we saw ODEs with discontinuous terms solved via Laplace transforms
- In Topic 4, we will learn how to use Fourier series to represent periodic functions that may have discontinuities

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Definition (Periodic extension of a function)

Let $f(x)$ be a function defined over $x \in (-L, L]$. The periodic extension of f is F:

$$\begin{aligned} F(x) &= f(x) \qquad \text{for} \qquad x \in (-L, L] \\ F(x + 2L) &= F(x) \end{aligned}$$

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Leaning outcomes:

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1. How Fourier series help model real-world applications

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2. How to construct Fourier series representations of continuous and discontinuous functions

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Leaning outcomes:

1. How Fourier series help model real-world applications
2. How to construct Fourier series representations of continuous and discontinuous functions
3. Be able to identify where a Fourier series converges and to which function

Lecture 15

We will:

1. Revise eigenvalue problems for real-symmetric matrices
2. Introduce eigenvalue problems for differential operators with periodic boundary conditions
3. Define an inner product and show orthogonality of an important set of eigenfunctions

Real and symmetric matrices (revision)

Let $\mathbf{A}$ be an $n \times n$ matrix and $\boldsymbol{v}$ a length n column vector, then

$$\mathbf{A}\,\boldsymbol{v} = \lambda\,\boldsymbol{v} \quad \text{is an eigenvalue problem}$$

If $\mathbf{A}$ is real-symmetric matrix:

1. The *eigenvalues*, λ_i, are real

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1. The *eigenvalues*, λ_i, are real
2. There are n linearly independent *eigenvectors*, $\boldsymbol{v}_i$, which are orthogonal

$$\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \boldsymbol{v}_i \cdot \boldsymbol{v}_j = 0 \quad \text{for} \quad i \neq j$$

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3. An arbitrary vector, $\boldsymbol{u}$, can be expressed as

$$\boldsymbol{u} = \sum_{i=1}^{n} c_i\,\boldsymbol{v}_i \qquad \text{where} \qquad c_i = \frac{\langle \boldsymbol{u}, \boldsymbol{v}_i \rangle}{\langle \boldsymbol{v}_i, \boldsymbol{v}_i \rangle} = \frac{\boldsymbol{u} \cdot \boldsymbol{v}_i}{|\boldsymbol{v}_i|^2}$$

Eigenvalues and eigenfunctions

Consider the second-order ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \lambda y = 0 \quad \Rightarrow \quad y = C_1 \sin\sqrt{\lambda}x + C_2 \cos\sqrt{\lambda}x \quad \text{if } \lambda > 0, \text{ etc}$$

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- λ is an *eigenvalue*
- $y(x)$ is an *eigenfunctions*
- Boundary conditions are needed to determine the eigenvalues and eigenfunctions

Exercise 4.1: Find the eigenvalues and eigenfunctions of

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2} \quad \text{with the periodic BCs} \quad y(-L) = y(L) \quad \text{and} \quad y'(-L) = y'(L)$$

Solve $-\frac{d^2}{dx^2}y = \lambda y$ for above BCs. let $\lambda \in \mathbb{R}$

Case $\lambda = 0$: $-\frac{d^2}{dx^2}y = 0 \Rightarrow y = C_1 + \cancel{C_2 x}$

$y(-L) = y(L)$: $\cancel{C_1} - C_2 L = \cancel{C_1} + C_2 L \Rightarrow 2C_2 L = 0 \Rightarrow C_2 = 0$

$y'(-L) = y'(L)$: $0 = 0$

So say

$y(x) = 1/2$

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For $\lambda = 0$

$$-\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = 0 \quad \Rightarrow \quad y = a\,x + b$$

Apply the periodic BCs

$$y(-L) = y(L) \quad \Rightarrow \quad -aL + b = aL + b \;\Rightarrow\; 2aL = 0 \;\Rightarrow a = 0$$

$$y'(-L) = y'(L) \quad \Rightarrow \quad b = b$$

Therefore $y(x) = b$ for any $b \neq 0$. We choose $b = 0.5$.

For $\lambda = k^2 > 0$ $\quad k \neq 0$

Solve $\quad -\frac{d^2}{dx^2} y = k^2 y \Rightarrow y(x) = C_1 \cos(kx) + C_2 \sin(kx)$

$y'(x) = -kC_1 \sin(kx) + kC_2 \cos(kx)$

$y(-L) = y(L)$: $\cancel{C_1 \cos(kL)} - C_2 \sin(kL) = \cancel{C_1 \cos(kL)} + C_2 \sin(kL)$

$\Rightarrow 2C_2 \sin(kL) = 0$

$y'(-L) = y'(L)$! $\quad kC_1 \sin(kL) + \cancel{kC_2 \cos(kL)} = -kC_1 \sin(kL) + \cancel{kC_2 \cos(kL)}$

$\Rightarrow 2kC_1 \sin(kL) = 0$

Therefore, choose $kL = n\pi$ for $n = 1, 2, 3, \ldots$

$y(x) = \cos(k_n x)$ and $y(x) = \sin(k_n x)$ where $k_n = \frac{n\pi}{L}$

$(n = 1, 2, 3, \ldots)$

For $\lambda = k^2 > 0$

$$-\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = k^2\, y \quad \Rightarrow \quad y(x) = C_1 \cos(k\,x) + C_2 \sin(k\,x)$$

Apply the periodic BCs

$$y(-L) = y(L) \quad \Rightarrow \quad C_1 \cos(kL) - C_2 \sin(kL) = C_1 \cos(kL) + C_2 \sin(kL)$$

$$\Rightarrow \quad 2\, C_2 \sin(kL) = 0$$

$$y'(-L) = y'(L) \quad \Rightarrow \quad k\, C_1 \sin(kL) + k\, C_2 \cos(kL) = -k\, C_1 \sin(kL) + k\, C_2 \cos(kL)$$

For both BCs, we require $\sin(kL) = 0 \Rightarrow k = k_n = n\,\pi/L$ for $n = 1, 2, 3, \ldots$.
As C_1 and C_2 are arbitrary, there are two sets of eigenfunctions

$$y = \cos(k_n x) \quad \text{and} \quad y = \sin(k_n x) \quad \text{for} \quad n = 1, 2, 3, \ldots$$

For $\lambda = -k^2 < 0$

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$$-\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = -k^2\, y \quad \Rightarrow \quad y(x) = C_1 \cosh(kx) + C_2 \sinh(kx)$$

The eigenfunction cannot be made to be periodic and hence there are no eigenfunction (details are an exercise).

Inner product

An inner product is needed to use the linear algebra framework.

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Definition

The inner product of functions f and g on $x \in (-L, L]$ is

$$\langle f, g \rangle = \frac{1}{L} \int_{-L}^{L} f g \, \mathrm{d}x$$

f and g are orthogonal if $\langle f, g \rangle = 0$

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Example

The inner product of x with x^3 is $\langle x, x^3 \rangle = \frac{1}{L} \int_{-L}^{L} x^4 \, dx = \frac{1}{L} \frac{1}{5} \left[x^5 \right]_{-L}^{L}$

$= \frac{1}{5L} 2L^5 = \frac{2L^4}{5}$

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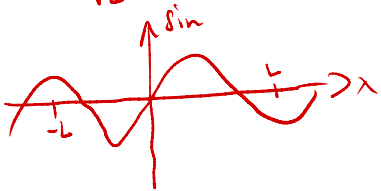
Exercise 4.2: Show orthogonality of eigenfunctions on previous slide

$y = 1/2$; $y = \cos(k_n x)$; $y = \sin(k_n x)$ where $k_n = n\pi/L$ $n = 1, 2, \ldots$

$$\langle 1/2, 1/2 \rangle = 1/L \int_{-L}^{L} 1/4 \, dx = 1/4L \left[x \right]_{-L}^{L} = 2L/4L = 1/2$$

$$\langle 1/2, \sin(k_n x) \rangle = 1/L \int_{-L}^{L} 1/2 \sin(k_n x) \, dx$$



$$= 0$$

$$\langle 1/2, \cos(k_n x) \rangle = 1/L \int_{-L}^{L} 1/2 \cos(k_n x) \, dx = 1/2L \left[1/k_n \sin(k_n x) \right]_{-L}^{L}$$

$$= 1/\cancel{2}Lk_n \times \cancel{2}\sin(k_n L) = 0 \quad \text{as } \sin(k_n L) = 0$$

Exercise 4.2: Show orthogonality of eigenfunctions on previous slide

$$\langle \cos(k_n x), \sin(k_m x) \rangle = 0$$

$$\langle \cos(k_n x), \cos(k_n x) \rangle = \frac{1}{L} \int_{-L}^{L} \cos^2(k_n x)\, dx$$

$$= \frac{1}{2L} \int_{-L}^{L} 1 + \cos 2k_n x \, dx$$

$$= \frac{1}{2L} \left[x + \cancel{\frac{1}{2k_n} \sin 2k_n x}^{\,0} \right]_{-L}^{L}$$

$$= 1$$

$$\cos 2X = \cos^2 X - \sin^2 X$$

$$= 2\cos^2 X - 1$$

$$\therefore \cos^2 X = \frac{1}{2}(1 + \cos 2X)$$

similarly $\langle \sin(k_n x), \sin(k_n x) \rangle = 1$

Exercise 4.2: Show orthogonality of eigenfunctions on previous slide

Ex.

$$\left.\begin{aligned} \langle \cos k_n x, \cos k_m x \rangle &= 0 \\ \langle \sin k_n x, \sin k_m x \rangle &= 0 \end{aligned}\right\} \; m \neq n$$

Exercise 4.2: Show orthogonality of eigenfunctions on previous slide

After the lecture ...

$$\langle \cos k_n x, \cos k_m x \rangle = \frac{1}{L}\int_{-L}^{L} \cos(k_n x)\cos(k_m x)\,dx$$

$n \neq m$

$$= \frac{1}{2L}\int_{-L}^{L} \cos(k_n + k_m)x + \cos(k_n - k_m)x \, dx$$

$$= \frac{1}{2L}\left[\frac{\sin(k_n+k_m)x}{k_n+k_m} + \frac{\sin(k_n-k_m)x}{k_n-k_m} \right]_{-L}^{L}$$

0 $\qquad$ 0

as $k_n + k_m = \frac{(n+m)\pi}{L}$ $\qquad$ as $k_n - k_m = \frac{(n-m)\pi}{L}$

$\cos X+Y = \cos X \cos Y - \sin X \sin Y$

$\cos X-Y = \cos X \cos Y + \sin X \sin Y$

$\therefore \cos X \cos Y = \frac{1}{2}\{\cos X+Y + \cos X-Y\}$

$\langle \sin k_n x, \sin k_m x \rangle$ is similar

Exercise 4.2: Show orthogonality of eigenfunctions on previous slide

First the easy cases, $\left\langle \frac{1}{2}, \frac{1}{2} \right\rangle = \frac{1}{L}\int_{-L}^{L} \frac{1}{4}\mathrm{d}x = \frac{1}{2}$. We have $\left\langle \frac{1}{2}, \cos k_n x \right\rangle = 0$ and $\left\langle \frac{1}{2}, \sin k_n x \right\rangle = 0$. We also have $\langle \cos k_n x, \sin k_n x \rangle = 0$ because the integrand will be an odd function.

Now the harder cases

$$\langle \cos k_n x, \cos k_n x \rangle = \frac{1}{L}\int_{-L}^{L} \cos^2 k_n x \mathrm{d}x = \frac{1}{2L}\int_{-L}^{L} 1 + \cos 2k_n x \mathrm{d}x = \frac{1}{2L}\int_{-L}^{L} \mathrm{d}x = 1$$

For $n \neq m$

$$\begin{aligned}\langle \cos k_m x, \cos k_n x \rangle &= \frac{1}{L}\int_{-L}^{L} \cos\frac{m\pi x}{L}\cos\frac{n\pi x}{L}\mathrm{d}x \\ &= \frac{1}{2L}\int_{-L}^{L} \cos\frac{(m+n)\pi x}{L} + \cos\frac{(m-n)\pi x}{L}\mathrm{d}x\end{aligned}$$

Now as $m \neq n$ we have that $\langle \cos k_m x, \cos k_n x \rangle = 0$.

We can similarly check that $\langle \sin k_n x, \sin k_n x \rangle = 1$ and $\langle \sin k_m x, \sin k_n x \rangle = 0$ for $m \neq n$ (left as a student exercise).